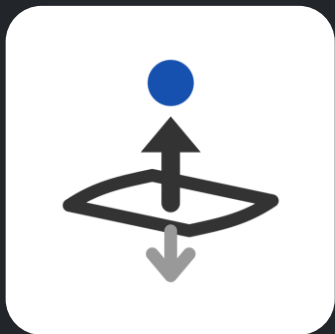




Orient to reference

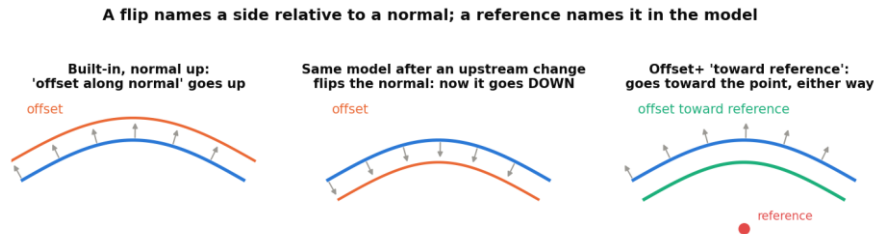
Flip surfaces so their normals point toward -- or away from -- a reference, so every feature downstream that follows the normal keeps working whatever happens upstream.



What it does

Many features pick their side by a surface's normal: Thicken's Thickness 1, Offset surface, extrude 'up to', Move face, Split front / back. Normals flip freely when an upstream loft, offset, mirror or sketch changes -- and the reference-side features keep their inputs' orientation. Orient to reference fixes the normal itself: each surface is turned to face the **Use it when** (or away).

- A built-in downstream follows a surface's normal
- Surfaces come from different sources and face different ways
- You want to SEE which way a set of surfaces faces (normal arrows)



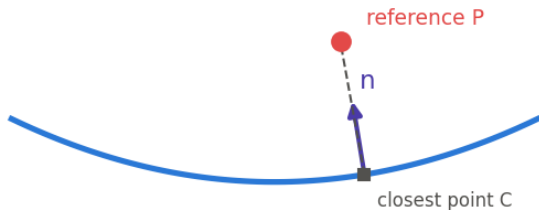
A flip names a side relative to a normal; a reference names it in the model.



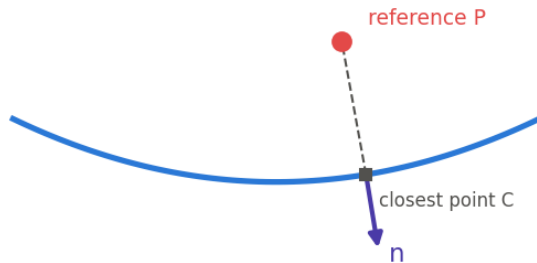
How the side is read

The side is read, not assumed: the sign of s says where the reference is for THIS normal

Normal up
 $s = (P - C) \cdot n = +31 \rightarrow$ reference on the FRONT



Normal down
 $s = (P - C) \cdot n = -31 \rightarrow$ reference on the BACK



Per surface body

$s = (P - C) \cdot n$ at the body's point nearest the reference. $s > 0$: the normal already points toward it.

Flip only what is wrong

Bodies facing the wrong way are flipped (opFlipOrientation); the rest are left exactly as they are.

Surfaces only

Solids always face outward and cannot be flipped; selecting one is an error.



The dialog

R1 Orient to reference, s...  

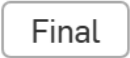
1 Surfaces to orient
Surface 18 

2 Reference
R1 reference point (1...  

4 ☒ Show normals

5 Arrow length

6 > Debug

#	Parameter	What it means
1	Surfaces to orient	Surface bodies, or faces of them (a face orients its whole surface body).
2	Reference	Geometry on the side the normals should face: best a mate connector or vertex near the surfaces.
3	Normals toward reference	The arrow. On: every normal points toward the reference. Off: away from it.
4	Show normals	While editing: arrows along each face's normal, after orientation. Green = already right, orange = flipped here.
5	Arrow length	Length of those arrows (10 mm).
6	Debug	Print sides: each surface's distance to the reference and whether it was flipped.



Examples: the normals, evaluated

Orient to reference on the test geometry (side view; normals evaluated in Onshape)

**R1, toward the reference: both sheets
already faced it -- left alone (green)**



● reference



**R2, the same sheets, AWAY from
the reference: both flipped (orange)**



● reference



**R3, half cylinder, reference on its axis,
toward: flipped to point at the axis**



Test geometry R1-R3, normals read back from Onshape after the feature ran

Two sheets with opposite normals and a reference between them: 'toward' leaves both alone because they already face it (green); 'away' flips both (orange). A half cylinder with the reference on its axis: every normal ends up pointing at the axis.



Outputs and messages

Published for Extract variables

Key	What it holds
output	Every selected surface, oriented.
flipped	The surfaces this feature flipped.
unchanged	The surfaces that already faced the right way.
flippedCount	How many were flipped.

Messages

	Message / what to do
Info	Flipped N of M surface(s); K already faced toward / away from the reference.
Error	A solid's faces cannot be flipped Select surfaces; solids always face outward.
Error	The reference lies on surface k Move the reference clearly off that surface.



Tips and limits



Put it right before the built-in

Orient the surfaces just before the Thicken, Offset surface or extrude that follows their normal.



Use a local reference

A mate connector or vertex near the surfaces, on the side meant. A far plane can read the wrong side where a surface curves.



Or skip the built-in

Thicken+ and Offset+ read the side from a reference directly and do not care which way the normal points.



One orientation per body

A merged surface (e.g. a Mutual Trim+ result) is one body and gets one orientation for all its faces.



Where it lives

Document	Reference_Side_Features (22764764a00a7f607dbc1c4d)
Tab	orient_to_reference (uses reference_side_utils)
Tests	Reference side tests studio: R1-R4 - devtools/onshape/check_reference_side_tests.py (34/34)
Explainer	docs/explainers/reference_side/reference_side_explained.md, sections 1.4 and 2.6
Related	Split+, Mutual Trim+, Offset+, Thicken+ (same reference rule)